Roll No. ....

TEC–801

B. TECH. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

COMPUTER ARCHITECTURE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1\. (a) What are various functional units of a basic computer ? Explain each in brief. (CO1, CO2)

OR

(b) Perform the following arithmetic operations using 8 bits in 2's complement form : (CO1, CO2)

(i) 16–24

(ii) 24–16

(iii) –24–16

2\. (a) Discuss 4 phases of an instruction cycle. (CO2)

OR

(b) Draw and explain the flowchart for interrupt cycle. (CO2)

P. T. O.

3. (a) Explain and classify the following instruction on the basis of memory reference, register reference and input and output reference instruction : (CO1, CO2)

   (i) ADD

   (ii) BSA

   (iii) CLA

   (iv) OUT

   (v) AND

OR

   (b) Differentiate between register reference and input output reference instruction code wit the help of an example. (CO1, CO2)

4. (a) Design a 4 bit synchronous binary counter. (CO2)

OR

   (b) Explain any *two* types of addressing modes with the help of an example. (CO2)

5. (a) Write short notes on RAM and ROM. (CO1, CO3)

OR

   (b) Design and explain 4 bit binary adder-subtractor circuit. (CO1, CO3)

Roll No. ....

TPE–014

B. TECH. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

TESTING FOR VLSI CIRCUITS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the challenges involved in testing VLSI circuits ? (CO1)

OR

(b) What is the purpose of testing VLSI circuits ? (CO1)

2. (a) How is functional testing different from structural testing for VLSI circuits ? (CO2)

OR

(b) Describe controllability and observability in detail. Write down the expressions for flip flop controllability and observability. (CO2)

3. (a) What are some commonly used fault models for VLSI circuit testing, and how do they differ ? (CO1)

P. T. O.

OR

(b) What is the role of test pattern generation in VLSI circuit testing and how is it performed ? (CO1)

4. (a) What is stuck-at fault testing and how is it performed ? (CO2)

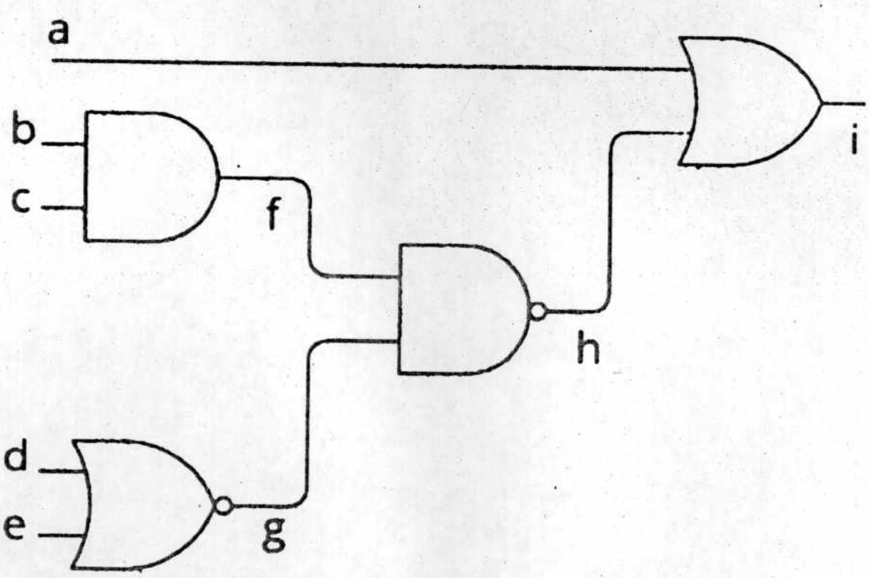
OR

(b) Define transistor faults. Explain transistor faults for NAND gate. (CO2)

5. (a) Explain the areas affecting VLSI testing techniques. (CO1)

OR

(b) Explain Boolean difference method of generating test vectors in detail. Generate the test vector considering stuck-at-0 fault at node "f" : (CO2)



Roll No. ....

TPE–015

B. TECH. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

WIRELESS SENSOR NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the properties and operating principle of wireless ad-hoc networks. Give *two* applications of wireless ad-hoc networks. (CO2)

OR

(b) Point at least *ten* differences between cellular networks and ad-hoc wireless networks. (CO2)

2. (a) What is wireless Sensor Networks (WSN) ? Explain its properties. How WSN is special class of wireless ad-hoc networks ? (CO1)

OR

(b) What are the unique challenges of Wireless Sensor Networks (WSN) ? Explain briefly. (CO1)

3. (a) Explain the applications of WSN in detail (at least *five*). (CO1/CO2)

OR

(b) What are the "wireless mesh networks" and "hybrid wireless networks". What are their advantages ? (CO1/CO2)

4. (a) Explain the use of Medium Access Control (MAC) protocols. What is the requirement and design constraint for wireless MAC protocol ? (CO3)

OR

(b) Explain the "Hidden Terminal Problem" and "Exposed Terminal Problem" in ad hoc wireless networks. (CO3)

5. (a) Explain the following in MAC protocols of WSN : (CO3

(i) Fixed Assignment protocols

(ii) Demand assignment protocols

OR

(b) Explain the classification of MAC protocol. How MACA (Multiple Access Collision Avoidance) protocol reduce the Hidden and Exposed Terminal Problems ? (CO3)

Roll No. ....

TPE–015

B. TECH. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

WIRELESS SENSOR NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the properties and operating principle of wireless ad-hoc networks. Give *two* applications of wireless ad-hoc networks. (CO2)

OR

(b) Point at least *ten* differences between cellular networks and ad-hoc wireless networks. (CO2)

2. (a) What is wireless Sensor Networks (WSN) ? Explain its properties. How WSN is special class of wireless ad-hoc networks ? (CO1)

OR

(b) What are the unique challenges of Wireless Sensor Networks (WSN) ? Explain briefly. (CO1)

3. (a) Explain the applications of WSN in detail (at least *five*). (CO1/CO2)

P. T. O.

OR

(b) What are the "wireless mesh networks" and "hybrid wireless networks". What are their advantages ? (CO1/CO2)

4. (a) Explain the use of Medium Access Control (MAC) protocols. What is the requirement and design constraint for wireless MAC protocol ?

(CO3)

OR

(b) Explain the "Hidden Terminal Problem" and "Exposed Terminal Problem" in ad hoc wireless networks. (CO3)

5. (a) Explain the following in MAC protocols of WSN : (CO3

(i) Fixed Assignment protocols

(ii) Demand assignment protocols

OR

(b) Explain the classification of MAC protocol. How MACA (Multiple Access Collision Avoidance) protocol reduce the Hidden and Exposed Terminal Problems ? (CO3)

Roll No. ....

TPE–016

B. TECH. (ECE) EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

CMOS ANALOG CIRCUIT DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Derive the drain current equation for a MOSFET and explain the I-V characteristics. (CO1)

OR

(b) What is channel length modulation ? Derive the drain current equation for an n-channel MOSFET including channel length modulation. (CO1)

2. (a) Explain the conditions in which MOSFET works as a constant current source. Design a voltage dependent current source of 1 mA with a MOSFET if, $\mu_n C_{ox} = 100\ \mu A/V^2$, $V_{TH} = 0.5$ V, W/L = 10. (CO1)

OR

(b) What is transconductance ? Explain the small signal operation of the MOSFET, when $\lambda = 0$. (CO1)

3. (a) Explain the working of a common source amplifier with a resistive load R_D with appropriate circuit diagrams and graphs. (CO2)

OR

(b) For a common source amplifier, $\mu_n C_{ox} = 100\ \mu A/V^2$, $V_{TH} = 0.5$ V, W/L = 10. The bias voltage $V_o = 1.2$ V and supply voltage $V_{DD} = 2$ V. Calculate the drain current if the MOSFET is in saturation. Also, find out the gain of this amplifier. (CO2)

4. (a) Explain the working of common source amplifier with current source load. Also, derive the gain equation as well. (CO2)

OR

(b) Explain the working of common source amplifier with diode connected load. Also, derive the gain equation as well. (CO2)

5. (a) What are the possible problems which may arise due to gain variation in CS amplifiers ? (CO1/CO2)

OR

(b) Explain the biasing conditions of a MOSFET and define cut-off, triode and saturation regions in detail. (CO1/CO2)

Roll No. ....

TPE–008

B. E. (ECE) (EIGHTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

NEURAL NETWORK AND MACHINE LEARNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the structure of a biological neuron and illustrate the model used for neural computing. (CO1)

OR

(b) Explain McCulloch-Pitts model of neural network and illustrate implementation of AND operation. (CO1)

2. (a) Design a pattern classifier using basic neural network model with S as a set of patterns to be classified into classes A and B : (CO2)

S = {000, 001, 010, 011, 100, 101, 110, 111}

A = {000, 001, 010, 100}

B = {011, 101, 110, 111}

P. T. O.

OR

(b) Illustrate implementation using neural network of the following table : (CO2)

Truth Table

x_1	x_2	x_1 AND (NOT x_2)
0	0	0
0	1	0
1	0	1
1	1	0

3. (a) Explain the structure of Perceptron. Implement OR Operation using the Perceptron. (CO2)

OR

(b) Explain the XOR Problem in Neural Network and suggest a solution. (CO2)

4. (a) What are the different types of learning schemes for neural network ? Explain with examples. (CO3)

OR

(b) With Hebb Learning Rule train the neural network to realise the following table : (CO3)

Training Set

	Input Patterns		Output
x_0	x_1	x_2	t
+ 1	1	1	1
+ 1	1	− 1	− 1
+ 1	− 1	1	− 1
+ 1	− 1	− 1	− 1

5. (a) With example, explain Perceptron Learning Rule. (CO3)

OR

(b) Explain any *two* activation functions. (CO3)